Solver Licensing, Technical Quality, and Architectural Strategy for the Riverline Poker Study Application

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Executive Summary and Analytical Framework

The development of Riverline—a local-first No-Limit Hold'em (NLHE) study application launching as a free tool with a prospective optional paid tier structured around subscriptions, supporter upgrades, paid cloud services, or user donations—presents a complex intersection of computational game theory and software intellectual property law. Because Riverline prioritizes deploying the highest-quality Game Theory Optimal (GTO) solver solutions strictly within legal boundaries, the architectural evaluation must distinguish between two fundamentally different modes of solver utilization. The first mode is offline data generation, or pre-computed strategy redistribution, wherein the developer executes solver binaries on private infrastructure offline, extracts the resulting numerical equilibrium approximations (specifically action frequencies, expected values, and range equities), and packages only those static or queryable data structures inside the Riverline application. The second mode is in-app code or binary inclusion, or embedded engine execution, wherein the developer links, compiles, or bundles the solver's source code or executable binary directly inside the distributed Riverline client to enable local on-device solving for end users.

Across the commercial and open-source poker solver ecosystem, licensing frameworks treat these two operational modes asymmetrically. Open-source copyleft licenses—specifically the GNU Affero General Public License v3.0 (AGPL-3.0) governing b-inary/postflop-solver and TexasSolver—impose strict viral source-disclosure obligations when solver code is compiled into a distributed application or exposed over a network server. However, under both mainstream open-source legal doctrine and explicit Free Software Foundation (FSF) guidance, the AGPL-3.0 does not reach numerical output data generated by running the program offline, provided the output does not contain copyrightable code or embedded artistic assets originating from the solver itself [Judgement]. Conversely, proprietary commercial solvers—including PioSOLVER, GTO+, Simple Postflop, HoldemResources Calculator (HRC), and MonkerSolver—govern usage through contractual End User License Agreements (EULAs) and Terms of Service (ToS) that regulate not only binary redistribution, but also whether the software may be operated for commercial purposes or used to populate third-party applications.

A comprehensive audit of actual license texts, EULAs, and benchmark data reveals an exceptionally clear, high-quality, and legally robust path for Riverline. For postflop heads-up play, b-inary/postflop-solver surpasses commercial incumbents such as PioSOLVER and GTO+ in raw Discounted Counterfactual Regret Minimization (DCFR) execution speed, matches their

equilibrium accuracy down to 0.1\% exploitability, and stands as the only engine in the world with exact multi-player card bunching support. For preflop, multiway, and tournament or Independent Chip Model (ICM) play, HoldemResources Calculator (HRC Pro) represents the premier actively maintained engine and uniquely includes an explicit commercial output redistribution clause in its v4.0+ EULA permitting licensees to *"distribute, openly share, or use the results of authorized calculations for commercial purposes"*.

Master Solver Comparison Matrix

The following table synthesizes the technical quality, output redistribution permissions across Riverline's four target distribution scenarios, code inclusion rules, commercial licensing availability, exact legal quotes with source URLs, and assessment confidence levels for all evaluated solvers. Within the matrix, Scenario (a) denotes a closed-source free application, Scenario (b) denotes a closed-source application with an optional paid tier, subscription, or donations, Scenario (c) denotes an open-source application under any Open Source Initiative (OSI) license distributed free of charge, and Scenario (d) denotes a paid tier, hosted service, or supporter content inside an open-source application.

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
b-inary / postflop-solver	**Top-tier HU Postflop.** DCFR (\gamma=3.0), SIMD/AVX2 & Apple Silicon Rust engine. Faster than PioSOLVER & GTO+; exact match at 0.1\% dEV.	**Yes** (AGPL does not cover numerical output) [Judgement]	**Yes** (AGPL does not cover numerical output) [Judgement]	**Yes** [Judgement]	**Yes** [Judgement]	**Allowed ONLY if Riverline is licensed under AGPL-3.0-or-later** (or GPLv3-linked per AGPL §13). Prohibits closed-source inclusion. Commercial sale of AGPL	**None available.** Author suspended OSS repo in Oct 2023 to build a business and no longer responds to issues/PRs; explicit permission given to fork under AGPL.	*"license = 'AGPL-3.0-or-later'"* (https://github.com/b-inary/postflop-solver/blob/main/Cargo.toml); *"As of October 2023, I have started developing a poker*	*High*

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	Exact 6-max bunching (up to 4 folded players). No preflop; no 3+ active postflop. **Suspended Oct 2023** (stable v0.1.0 lib / v0.2.7 app).					app is permitted if full source is provided.		*solver as a business and have decided to suspend development..."* (https://github.com/b-inary/postflop-solver/issues/46).	
TexasSolver (bupticybee)	**Fast C++ / GPU HU Postflop, Accuracy Caveats.** Supports HU NLHE & Short Deck, node locking, JSON export, GPU acceleration.	**Yes** (under standard AGPLv3 output rules) [Judgement]	**Yes** (under AGPLv3, though author claims commercial service requires license) [Judgement]	**Yes** [Judgement]	**Yes** [Judgement]	**Open Source (AGPL-3.0)** if code is used under AGPLv3. README claims binary integration is allowed, but source integration or internet service	**Available via author** (icybee@yeah.net). Dual-licensing is the project's primary revenue model; recipients listed in licensed_list.txt. Price not public.	*"If you integrate the release package (binary) into your software, Yes, you can do that. If you want to integrate the code of the solver into your software or*	**Medium-High** (README contradicts AGPLv3)

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	Independent benchmark by b-inary flagged divergent/unverifiable EV outputs vs Pio/GTO+. No preflop/multiway. **Actively maintained (2026)**.					requires a commercial license (licensed_list.txt).		*provide service through internet, then you need to contact me for a commercial license"* (https://github.com/bupticybee/TexasSolver).	
HRC (Holdem Resources Calculator v4+)	**Gold Standard Preflop & Multiway.** Supports multiway preflop (ChipEV, ICM, PKO, Rake) and postflop realization up to 16,384	**Yes** (Explicitly permitted by EULA)	**Yes** (Explicitly permitted for commercial purposes)	**Yes** (Explicitly permitted: "openly share")	**Yes** (Explicitly permitted for commercial purposes)	**Strictly Prohibited.** Binary-only proprietary software. Cannot distribute binary, share keys, scrape UI/memory, or integrate into a web	**Included in Standard License for Outputs!** HRC Classic: $119.90/yr; **HRC Pro: $359.90/yr** ($29.99/mo annual, $39.99/mo	*"However, the Licensee is permitted to distribute, openly share, or use the results of authorized calculations for commercial purposes*	**Very High**

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	buckets (HRC Pro), unlimited tree size, scripting. **Actively maintained (v4.0+)**.					service backend.	quarterly). Backend/engine licensing not offered.	*provided these results were produced by the Licensee in compliance with this Agreement and applicable law."* (https://www.holdemresources.net/legal/eul[span_21](start_span)[span_21](end_span)[span_26](start_span)[span_26](end_span)a/hrc_v4).	
PioSOLVER (v3.x)	**Industry Benchmark HU Solver.** Exact HU postflop	**Grey / Risky** (Allowed for saves/coaching, barred if	**No / High Risk** (Without Business License, queryabl	**Grey / Risky** (Same "on-demand service" restrictio	**No / High Risk** (Requires Business License)	**Binaries Prohibited.** Cannot distribute PioSOLVER/PioVi	**Available: €5,000 / month** (covers 100 floating	*"Common sense: you can share results/saves/observations*	**High**

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	(and HU preflop in Edge), UPI scripting, node-locking, ICM, aggregation reports. No multiway (3+ players). **Actively maintained**.	deemed an "on-demand service") [Judgement]	e app database risks termination) [Judgement]	n applies) [Judgement]	[Judgement]	ewer binaries. Can integrate UPI wrapper *only* if end user owns a licensed copy of PioSOLVER (or attributes Pio Free).	PCs, includes Linux engine, allows on-the-fly compute, API sales, and cloud offerings). Standard Pro is ~€300 once.	*; you can use it to produce coaching materials, books, videos etc. you can't turn into on-demand service where specific PioSOLVER results would be available without actually purchasing PioSOLVER program."* (https://piosolver.com/docs/licensing/).	
GTO+ (StoxEV B.V.)	*High-Value HU Postflop.* Fast converge	**No / Breach Risk** (EULA limits	**No** (Strictly breaches "personal,	**No / Breach Risk** (Public distributi	**No** (Strictly breaches "personal,	**Strictly Prohibited.** Proprietary binary	**Not publicly offered.** Standard license is	*"The license that is granted to You*	**High**

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	nce to 0% Nash distance, tiny savefiles (hundreds of KB), database scripting, turn/river aggregate reports. HU postflop only. **Actively maintained (v182)**.	use to "personal, non-commercial") [Judgement]	non-commercial" EULA) [Judgement]	on exceeds personal use) [Judgement]	non-commercial" EULA) [Judgement]	($75). EULA forbids licensing, distributing, hosting, or making the Application available to any third party.	$75 personal only. Requires bespoke written waiver from StoxEV B.V. (help@gtoplus.com).	*by the Company is solely for your personal, non-commercial purposes strictly in accordance with the terms of this Agreement."* (https://www.gtoplus.com/eula/).	
Simple Postflop (SimplePoker)	**Fast HU & Suite Ecosystem.** Standalone ($299) has 3 algorithms (A1 compressed, A2 narrow, A3 fast) + preflop; separate	**No / High Risk** (ToS bars exploiting "use of the Service" without written consent) [Judgement]	**No** (Requires express written permission) [Judgement]	**No / High Risk** (Requires express written permission) [Judgement]	**No** (Requires express written permission) [Judgement]	**Strictly Prohibited.** Single/dual-PC hardware-locked proprietary binaries. Cannot reproduce, copy, or resell any portion of the	**Bespoke B2B only.** SimplePoker sells its own competing apps (GTOBase, Simple GTO Trainer) and licenses to select	*"You agree not to reproduce, duplicate, copy, sell, resell or exploit any portion of the Service, use of the Service,*	**High**

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	Simple 3-Way & Simple Preflop Holdem. **Maintained**.					Service.	partners (e.g., GrinderLab) via written agreement.	*or access to the Service... without express written permission by us."* (https://simplepoker.com/terms_and_conds).	
Monker Solver (Monker Ware)	**Heavy Multiway & PLO Solver.** Supports multiway NLHE & PLO preflop/postflop via abstraction. Requires massive RAM (200GB+); noisier MCCFR convergence than HRC Pro.	**Yes** (ToS does not restrict locally computed outputs; wide precedent) [Judgement]	**Yes** (Only server-side "Purchased ranges" are restricted; wide precedent) [Judgement]	**Yes** [Judgement]	**Yes** [Judgement]	**Strictly Prohibited.** Proprietary Java application (€499). Accounts are personal and non-transferable.	**Standard €499 one-time license** contains no prohibition on redistributing user-computed ranges (unlike Monker Ware's own server-side purchased ranges).	*"Monker Ware accounts are personal and non-transferable... Purchased ranges are stored server-side to deter piracy and allow us to make improvements."*	**Medium-High**

Solver	Quality Summary (Accuracy, Speed, Coverage, Maintenance)	Outputs in (a) Closed Free	Outputs in (b) Closed Paid	Outputs in (c) Open Free	Outputs in (d) Open Paid	Code / Binary Inclusion Terms	Commercial / Redistribution Licence	Source Quote & Official Link	Confidence
	Sparsely maintained.							(https://monkerware.com/tos.html).	
OpenSpiel (DeepMind)	**Research Reference Library.** Contains reference Python/C++ implementations of DCFR, Linear CFR, CFR+, Deep CFR. Not an optimized full NLHE production solver out-of-the-box. **Maintained**.	**Yes** (Apache-2.0) [Judgement]	**Yes** (Apache-2.0) [Judgement]	**Yes** (Apache-2.0) [Judgement]	**Yes** (Apache-2.0) [Judgement]	**Permitted in BOTH Closed and Open Source.** Apache-2.0 allows commercial closed-source modification and distribution with only attribution and notice retention [Judgement].	**Free under Apache-2.0.** No royalties or source-disclosure obligations for derivative works [Judgement].	*"This implements Discounted CFR and Linear CFR, from Noam Brown and Tuomas Sandholm..."* (https://github.com/deepmind/open_spiel/blob/master/open_spiel/python/algorithms/discounted_cfr.py).	**High**

Cross-Cutting Legal Doctrine: Does Copyleft (AGPL-3.0) Reach Solver Output Data?

Before examining individual engines, a foundational legal question must be resolved for

Riverline's primary architectural plan of running solvers offline and shipping only strategy frequencies and expected values (EVs) inside the application: whether the GNU Affero General Public License v3.0 (AGPL-3.0) infects the numerical outputs generated by running an AGPL-licensed solver.

The Free Software Foundation (FSF) and Mainstream Copyright Doctrine

Under copyright law, a software license such as the GPLv3 or AGPLv3 derives its legal force entirely from copyright ownership over the underlying software code. Consequently, the copyleft obligations of AGPL-3.0 apply only to the original program, modified versions of the program, or derivative works based on the program. The Free Software Foundation, which drafted and stewards both the GPLv3 and AGPLv3, addresses this exact question in its official FAQ under the heading *"In what cases is the output of a GPL program covered by the GPL too?"* (https://www.gnu.org/licenses/gpl-faq.html#WhatCaseIsOutputGPL). The FSF establishes the general rule that the output of a program is not covered by the copyright on the code of the program, and therefore the GPL or AGPL does not apply to the program's output—regardless of whether that output is piped into a file, captured in a screenshot, or bundled into another software product. Furthermore, the FSF explicitly prohibits copyright holders from adding terms to the GPL that would restrict or require citation for research papers or data produced using a GPL program's output (https://www.gnu.org/licenses/gpl-faq.html#RequireCitation).
According to the FSF, a program's output becomes subject to the GPL or AGPL only in the narrow exception where the program copies substantial, copyrightable portions of its own code, full-screen text, artwork, or audio assets directly into the output file. Classic examples include a parser generator like GNU Bison injecting a copyrighted C skeleton file into the generated parser—which necessitated a special Bison exception—or a video game outputting its own copyrighted music and graphical sprites. Where the output is purely the result of data processing on user-supplied parameters without copying the program's creative code or assets, the output belongs to the user who ran the program and may be licensed under any terms they choose [Judgement].

Application to GTO Poker Solver Outputs

When Riverline executes b-inary/postflop-solver or TexasSolver offline, the developer supplies pure mathematical inputs: initial hand ranges consisting of 1,326 hole-card combination weights, stack-to-pot ratios, allowed bet-size trees, and community board cards. The solver executes iterative Counterfactual Regret Minimization arithmetic in memory and outputs floating-point arrays representing action probabilities and expected values.
This mathematical process avoids copyleft infection on three distinct legal grounds [Judgement]. First, there is no code injection: neither b-inary/postflop-solver nor TexasSolver copies its Rust or C++ source code, user interface artwork, or creative text into the exported JSON, CSV, or binary strategy arrays [Judgement]. Even if Riverline utilizes b-inary/postflop-solver's native bincode serialization (save_data_[span_6](start_span)[span_6](end_span)to_file) to dump the internal PostFlopGame struct, that file contains only serialized numerical node states and tree topology rather than executable code. Nevertheless, converting those numerical outputs into Riverline's own clean serialization schema—such as custom flatbuffers, SQLite tables, or compressed JSON—eliminates even the slightest structural dependency on PostFlopGame

struct layouts [Judgement].
Second, mathematical equilibria are non-copyrightable facts. Under United States copyright law (17\text{ U.S.C. \S } 102(\text{b}) and *Feist Publications, Inc. v. Rural Telephone Service Co.*) as well as European Union copyright directives, mathematical truths, discoveries, and algorithmic outputs dictated by functional efficiency lack original creative expression and cannot be copyrighted [Judgement]. Because independent solvers (b-inary/postflop-solver, PioSOLVER, and GTO+) given identical game trees converge to the exact same Nash equilibrium strategies and expected values, the resulting numbers are mathematical facts inherent to the game tree rather than creative works of the solver programmer [Judgement].
Third, under European Union and United Kingdom *sui generis* database law (Directive 96/9/EC), a non-original collection of data may still be protected by a database right if there has been a substantial investment in obtaining, verifying, or presenting the contents [Judgement]. Crucially, when Riverline's developer runs an open-source solver on Riverline's own hardware, Riverline is the entity making the substantial investment in CPU and GPU compute time, electricity, and tree curation to generate and verify the database. Consequently, any *sui generis* database right vests in Riverline rather than in the author of the open-source solver software [Judgement].

Technical and Legal Deep-Dive by Solver

To ground the architectural and legal evaluation of each individual solver, the visual dashboard below synthesizes the benchmark speed, convergence verification, game-tree coverage, and licensing boundaries across Riverline's target deployment models.

1. b-inary/postflop-solver (and desktop-postflop / wasm-postflop)

Developed by Wataru Inariba (b-inary), postflop-solver is an open-source postflop solver library written in Rust, accompanied by a Tauri native desktop application (desktop-postflop) and a browser-based WebAssembly port (wasm-postflop). The engine implements the state-of-the-art Discounted Counterfactual Regret Minimization (DCFR) algorithm introduced by Brown and Sandholm (2018), modified with a discounting parameter of \gamma = 3.0 rather than the original paper's 2.0, alongside cumulative strategy resets whenever the iteration count reaches a power of 4, which completely eliminates weak actions from the output strategy. To maximize throughput, the developer manually inspected compiler assembly outputs, utilized unsafe Rust in critical hot loops, and enforced SIMD vectorization (AVX2 on x86-64 Haswell/Zen+ processors and native instructions on Apple Silicon M1+) alongside rayon multithreading and an optional nightly custom memory allocator (custom-alloc). In controlled benchmarks on an AMD Ryzen 7 3700X (16 threads) using PioSOLVER's "3betpotFAST" preset solved down to 0.1\% exploitability, native postflop-solver (desktop-postflop) outperformed both paid PioSOLVER and GTO+ in execution time.
In terms of numerical precision and coverage, b-inary/postflop-solver uses 32-bit floating-point numbers (f32) in most calculations while accumulating temporary summations in 64-bit floating-point precision (f64), and includes an optional memory compression mode storing values as 16-bit integers (i16) with a single 32-bit scaling factor per node. The engine performs zero card abstraction, combining only exact isomorphic turn and river runouts, such as collapsing the three non-dealt suits on a monotone flop. Benchmark comparisons confirm that postflop-solver, PioSOLVER, and GTO+ return nearly identical strategies, equities, and EVs. It supports heads-up postflop No-Limit Hold'em and Fixed-Limit Hold'em (FLHE), raked pots,

geometric bet sizing, donk bet trees, and engine-level node locking. Crucially, it is the first and only postflop solver in the world to implement exact card bunching, supporting up to four folded players in a 6-max game by counting exact card combinations rather than manipulating deck probability distributions heuristically, although enabling bunching significantly increases terminal node evaluation time. The library does not support preflop solving or postflop spots with three or more active players. Regarding maintenance, on October 1, 2023, Wataru Inariba suspended development of postflop-solver, desktop-postflop, and wasm-postflop (leaving v0.2.7 as the final application release) to develop a commercial poker solver as a business. Despite this suspension, the library is functionally complete for heads-up postflop play, with engine-side node locking and save/load already implemented, exhibiting only minor dependency aging in bincode = "2.0.0-rc.3" that community developers easily bypass or patch in Rust.
With respect to output redistribution across Scenarios (a), (b), (c), and (d), the repository's Cargo.toml (https://github.com/b-inary/postflop-solver/blob/main/Cargo.toml) specifies *"license = 'AGPL-3.0-or-later'"*, and the README (https://github.com/b-inary/postflop-solver) contains no extra restrictions on output or usage, stating only: *"See the GNU Affero General Public License for more details."*. Because AGPL-3.0 does not extend to numerical outputs generated offline, redistributing strategies, frequencies, and EVs computed by b-inary/postflop-solver is legally permitted in all four scenarios—(a) Closed Free, (b) Closed Paid, (c) Open Free, and (d) Open Paid [Judgement].
Conversely, including b-inary/postflop-solver source code as a Rust crate dependency (postflop-solver = { git = "https://github.com/b-inary/postflop-solver" }) inside Riverline is strictly prohibited in a closed-source application. Code inclusion is permitted in an open-source application only if the entire combined Riverline application is licensed under AGPL-3.0-or-later (or GPLv3 when linked under AGPLv3 Section 13, which retains AGPLv3 network source-disclosure obligations) [Judgement]. Under AGPL-3.0, Riverline may still charge money for the application or offer a paid tier, as the GPL and AGPL explicitly permit selling copies or commercial services (https://www.gnu.org/licenses/gpl-faq.html), provided that every user who downloads the client or interacts with it over a network receives immediate access to the complete Corresponding Source Code of the entire application. Separate commercial licensing is unavailable; in Issue #46 (https://github.com/b-inary/postflop-solver/issues/46), b-inary announced: *"Therefore, the following repositories will no longer be updated and I will no longer respond to issues and pull requests."*. The author explicitly encouraged developers to fork, modify, and redistribute the project under the AGPL license. Legal risk for offline data shipping is exceptionally low, though if Riverline embeds the solver code under AGPL-3.0, legal counsel should confirm that any proprietary backend billing or authentication servers remain separated via clean API boundaries so they do not constitute a single combined derivative work [Judgement].

2. TexasSolver (bupticybee/TexasSolver and GPU Edition)

Created by bupticybee, TexasSolver is an open-source C++ (with a Qt 6.2.1 GUI) and GPU-accelerated GTO solver for Heads-Up postflop Texas Hold'em and Short Deck. It runs on Windows, macOS, and Linux, and supports node locking, batch solving, cross-language calls, and direct strategy dumps to JSON files. A free GPU-accelerated edition (https://bupticybee.github.io/texassolvergpupage/) provides substantial speedups over CPU solvers, and the repository remains actively maintained through 2026 with over 2,600 GitHub stars and 470 forks. However, on the critical criterion of accuracy, b-inary tested TexasSolver (v0.2.0) against PioSOLVER, GTO+, and postflop-solver on the "3betpotFAST" benchmark

(https://github.com/b-inary/wasm-postflop) and reported: *"We can see that WASM Postflop, PioSOLVER, and GTO+ return nearly identical results... TexasSolver returned a different solution, which is presumably incorrect. Either way, we cannot verify the correctness because we cannot see the overall EV in TexasSolver."*. Like b-inary/postflop-solver, TexasSolver does not solve preflop or multiway postflop spots.
The licensing posture of TexasSolver is complicated by contradictory statements written by its author. While the repository (https://github.com/bupticybee/TexasSolver) is formally licensed under GNU AGPL v3, its README adds custom Q&A clauses and references a file named licensed_list.txt naming commercially licensed entities:
"Q: Is the solver really free? A: Yes, for personal users, the solver is completely opensourced and free. Q: Can I upload it to other websites or forums? Can I share it with my friend? A: No, you can only put the link of this project to other website, not the binary, this project is under AGPL-V3 license, and these kind of actions violates this license. Q: Can I integrate it to my software? A: If you integrate the release package (binary) into your software, Yes, you can do that. If you want to integrate the code of the solver into your software or provide service through internet, then you need to contact me for a commercial license, which is also the main profit-making method of this project."
With respect to output redistribution across Scenarios (a), (b), (c), and (d), neither the AGPLv3 license nor the custom README Q&A restricts the numerical JSON strategy files produced by running TexasSolver offline, making output redistribution legally permissible across all four scenarios [Judgement]. However, for code or binary inclusion, the author's README makes legally incoherent claims about AGPLv3—asserting that sharing binaries with a friend violates AGPLv3 (whereas AGPLv3 expressly grants the right to distribute binaries alongside source code), while simultaneously claiming users may integrate the release binary into their own software, yet requiring a paid commercial license (icybee@yeah.net, recorded in licensed_list.txt) to integrate the source code or provide an internet service. A lawyer would need to evaluate whether the README Q&A constitutes an invalid "further restriction" that recipients may legally strip under AGPLv3 Section 7, or whether the author's conflicting intent creates an estoppel or litigation nuisance if used without purchasing a commercial license [Judgement]. Given TexasSolver's documented benchmark accuracy divergence from PioSOLVER and GTO+, combined with b[span_250](start_span)[span_250](end_span)[span_252](start_span)[span_252](end_span)-inary/postflop-solver's superior accuracy and clean AGPL-3.0 licensing, TexasSolver is unnecessary for Riverline [Judgement].

3. HoldemResources Calculator (HRC v4.0+ / HRC Pro)

Developed by HoldemResources GmbH in Vienna, Austria, HoldemResources Calculator—and specifically its flagship tier, HRC Pro in version 4.0 and higher—is the premier commercial solver for preflop, multiway, and tournament or cash game preflop-to-postflop calculations. HRC Pro solves full multiway preflop trees at any stack depth, covering 100bb+ raked cash games, multi-table tournament (MTT) ICM, Progressive Knockout (PKO) bounties, and ChipEV formats with unlimited tree sizes constrained only by system RAM. Unlike older preflop tools that rely on static equity approximations, HRC Pro models full postflop equity realization using up to 16,384 buckets, and supports adding actions after tree creation, scripted tree building, and dedicated postflop solves (whereas the entry-level HRC Classic tier is capped at 50,000 nodes and 256 postflop buckets for stacks up to 30 big blinds). The software is actively maintained, with its v4.0+ EULA updated in October 2024 and continuous feature development.

Regarding output redistribution in Scenarios (a), (b), (c), and (d), the HRC v4.0+ EULA (https://www.holdemresources.net/legal/eula/hrc_v4) contains the clearest, most explicit commercial output permission in the poker software industry:
"Sharing the license key or making the Product accessible to others is prohibited, especially through automated means. This includes integrating the Product into the backend of a web service to allow others to submit calculations or use the Product indirectly. However, the Licensee is permitted to distribute, openly share, or use the results of authorized calculations for commercial purposes, provided these results were produced by the Licensee in compliance with this Agreement and applicable law."
The EULA further specifies the required method of data extraction:
"Automated scraping of information from the Product's user interface or memory is expressly prohibited. Results should only be exported using the export and save options provided within the Product."
Consequently, shipping HRC Pro outputs is explicitly permitted in all four scenarios: (a) Closed Free, (b) Closed Paid, (c) Open Free ("openly share"), and (d) Open Paid ("for commercial purposes"). The sole compliance requirements are that the Riverline developer holds an active, valid HRC license key, executes calculations through the official launcher, and exports the resulting strategy data strictly via HRC's built-in export and save functions rather than memory or UI scraping. Conversely, code or binary inclusion inside Riverline is strictly prohibited; HRC is distributed in unmodified binary form only, source code is not provided, and integrating HRC into a web service backend is expressly forbidden. Because commercial output redistribution is granted directly within the standard EULA, Riverline requires only a standard single-user HRC Pro license, which is publicly priced at $359.90 per year$ (29.99/month billed annually, or $119.97 billed quarterly, permitting two active installations) (https://www.holdemresources.net/hrc/pricing). Legal risk for offline data shipping is near zero, though counsel should note that the EULA is governed by Austrian law with exclusive jurisdiction in the Vienna Commercial Court [Judgement].

4. Pio[span_75](start_span)[span_75](end_span)SOLVER (PIOSOFTWARE LTD)

Sold by PIOSOFTWARE LTD (registered in Cyprus), PioSOLVER (currently in v3.x) is the foundational commercial benchmark solver for Heads-Up NLHE postflop play, with heads-up preflop solving available in the high-end PioSOLVER Edge edition. It provides rapid convergence, full Universal Poker Interface (UPI) command-line scripting, node locking, ICM calculations, rake modeling, and multi-flop aggregation reports, though it does not support multiway (3+ player) preflop or postflop solving. The software is actively maintained under a perpetual license model that includes one year of feature updates for v3 purchases.
With respect to output redistribution across Scenarios (a), (b), (c), and (d), the official PioSOLVER Licensing & Terms page (https://piosolver.com/docs/licensing/) states:
"Common sense: you can share results/saves/observations; you can use it to produce coaching materials, books, videos etc. you can't turn into on-demand service where specific PioSOLVER results would be available without actually purchasing PioSOLVER program... We reserve right to interpret those terms and terminate your license if the terms are not upheld."
While sharing individual save files or static observations in books and coaching videos is permitted, packaging a queryable database of PioSOLVER solutions inside a software application where end users look up specific board and spot strategies on demand directly

implicates the prohibition against turning outputs into an *"on-demand service where specific PioSOLVER results would be available without actually purchasing PioSOLVER program"* [Judgement]. Even in a free application under Scenarios (a) or (c), if Riverline functions as a substitute for purchasing PioSOLVER by serving specific node frequencies on demand, PIOSOFTWARE LTD explicitly reserves the unilateral right to interpret those terms and terminate the developer's license [Judgement]. In paid-tier Scenarios (b) and (d), doing so without a commercial Business License represents a clear contractual breach [Judgement]. Regarding binary or code inclusion, the licensing terms stipulate: *"You can not distribute PioSOLVER or PioViewer binaries. You can integrate PioSOLVER with your software as long as using this functionality requires licensed PioSOLVER. If you wish to integrate PioSOLVER free version in your software then you only need to give information about the fact (link to our website would be nice as well)."* (https://piosolver.com/docs/licensing/). Thus, Riverline may legally include a UPI bridge that connects to an end user's own locally installed, licensed copy of PioSOLVER, but cannot bundle PioSOLVER Pro or Edge binaries inside Riverline. For commercial data services and backend solving, the PioSOLVER Business FAQ (https://piosolver.com/docs/faq/business/) details a dedicated Business License priced at €5,000 per month covering 100 floating computers and including a native Linux engine. This enterprise tier explicitly permits *"computing on the fly, selling API access, developing cloud offerings, and other commercial/business purposes"*. At €60,000 per year, however, this license is economically unjustifiable for Riverline when b-inary/postflop-solver produces identical heads-up postflop equilibria at zero cost [Judgement].

5. GTO+ (StoxEV B.V. / Scylla)

Developed by StoxEV B.V. in Purmerend, Netherlands (led by developer "Scylla"), GTO+ is a widely used $75 commercial Heads-Up postflop NLHE solver. It converges all the way down to a 0\% Nash distance, features low memory overhead and ultra-compact savefiles measuring only a few hundred kilobytes (by recalculating turn and river streets on the fly), and includes database scripting, turn and river aggregate reports, equity buckets, and MTT ICM conversions. It is strictly a postflop heads-up solver (https://www.gtoplus.com/: *"calculating Nash-equilibrium strategies in postflop heads-up games"*) and is actively maintained, with v182 released in October 2025.

Despite its low $75 retail price, GTO+ has the most restrictive EULA among all evaluated solvers regarding output usage. The official GTO+ End-User License Agreement (https://www.gtoplus.com/eula/, last updated May 17, 2023) states:

"Scope of License: The Company grants You a revocable, non-exclusive, non-transferable, limited license to download, install and use the Application strictly in accordance with the terms of this Agreement. The license that is granted to You by the Company is solely for your personal, non-commercial purposes strictly in accordance with the terms of this Agreement. License Restrictions: You agree not to, and You will not permit others to: License, sell, rent, lease, assign, distribute, transmit, host, outsource, disclose or otherwise commercially exploit the Application or make the Application available to any third party. Copy or use the Application for any purpose other than as permitted under the above section 'License'."

Because the EULA explicitly prohibits *using* the Application for any purpose other than *"your personal, non-commercial purposes"*, running GTO+ to generate strategy databases for a monetized application in Scenarios (b) and (d) is a direct breach of contract [Judgement]. Furthermore, even in free-tier Scenarios (a) and (c), distributing a public software application populated with GTO+ databases to third parties exceeds personal use and carries a high risk of

EULA termination and breach claims [Judgement]. Binary inclusion is likewise strictly prohibited, and StoxEV B.V. offers no public commercial license tier; utilizing GTO+ for Riverline would require negotiating a bespoke written commercial waiver via help@gtoplus.com or support@gtoplus.com [Judgement].

6. Simple Postflop (and SimplePoker Suite: Simple Preflop Holdem, Simple 3-Way)

Operated by Simple Software Solutions FZE LLC, the SimplePoker ecosystem comprises Simple Postflop (a $299 standalone perpetual desktop license for two PCs covering HU postflop and preflop with three selectable algorithms: A1 with 4x RAM compression, A2 for narrow ranges, and A3 for ultra-fast convergence), Simple 3-Way (a subscription 3-player postflop solver), and Simple Preflop Holdem (a multiway preflop solver supporting ChipEV, ICM, and cash rake via advanced bucketing).
From a legal standpoint, the SimplePoker Terms and Conditions (https://simplepoker.com/terms_and_conds, Section 2 – General Conditions) impose a broad prohibition on commercial or third-party exploitation:
"You agree not to reproduce, duplicate, copy, sell, resell or exploit any portion of the Service, use of the Service, or access to the Service or any contact on the website through which the service is provided, without express written permission by us."
Because Simple Software Solutions FZE LLC explicitly forbids exploiting *"use of the Service"* without express written permission—and actively monetizes its own competing study and training applications (GTOBase, Simple GTO Trainer, and GTO Sensei) alongside formal B2B licensing partnerships (such as GrinderLab)—shipping Simple Postflop, Simple 3-Way, or Simple Preflop Holdem outputs inside Riverline without a signed B2B contract is prohibited across Scenarios (a), (b), (c), and (d) [Judgement]. Binary inclusion is also strictly prohibited, as licenses are hardware-locked per PC. Any commercial redistribution arrangement requires direct written negotiation with support@simplepoker.com.

7. MonkerSolver (MonkerWare)

MonkerSolver (€499 one-time license) is a cross-platform Java solver capable of solving multiway NLHE and Pot-Limit Omaha (PLO4, PLO5, and PLO6) preflop and postflop trees using abstraction and Monte Carlo CFR variants. For multiway NLHE preflop study, however, MonkerSolver requires massive hardware resources (often 200GB+ of RAM for full 6-max trees), takes days to converge, and exhibits higher strategy noise on low-frequency combinations compared to HRC Pro's deterministic 16,384-bucket postflop realization engine [Judgement]. Moreover, its maintenance and support are sparse compared to HRC.
The official MonkerWare Terms of Service (https://monkerware.com/tos.html) are remarkably brief and state:
"The following terms are applied to all services provided by MonkerWare (including but not limited by MonkerViewer and MonkerSolver)... MonkerWare accounts are personal and non-transferable. Sharing accounts may result in a ban from the service... Purchased ranges are stored server-side to deter piracy and allow us to make improvements. Purchased ranges may therefore change over time."
Notably, the MonkerWare Terms of Service contain no restriction on commercial use or redistribution of ranges computed locally by a user inside MonkerSolver; the anti-piracy

restriction applies exclusively to pre-solved *"Purchased ranges"* bought from MonkerWare's server-side range store. Consequently, redistributing locally computed MonkerSolver outputs is legally permitted across Scenarios (a), (b), (c), and (d), though binary inclusion of the proprietary Java solver remains prohibited [Judgement]. For NLHE specifically, however, HRC Pro is both technically superior in convergence quality and backed by an affirmative EULA grant rather than mere contractual silence [Judgement].

8. Other Solvers and Data Providers Worth Considering

Beyond the seven primary engines, several adjacent libraries and commercial platforms merit evaluation. DeepMind's OpenSpiel (https://github.com[span_149](start_span)[span_149](end_span)/deepmind/open_spiel) is an open-source framework licensed under the permissive Apache-2.0 license that includes clean C++ and Python reference implementations of Discounted CFR (discounted_cfr.py), Linear CFR, CFR+, MCCFR, and Deep CFR. While its generic game-tree representation lacks the Hold'em-specific SIMD vectorization and isomorphism compression of b-inary/postflop-solver, its Apache-2.0 license permits both source-code embedding and output distribution in closed-source and open-source commercial applications alike [Judgement].
On the proprietary cloud side, platforms such as GTO Wizard (and GTO Wizard AI), Deepsolver, Ollin Solver, and NashPoint illustrate modern solver architecture but are not viable legal data sources for third-party redistribution. GTO Wizard initially populated its 10,000,000+ pre-solved spot library using commercial solvers before engineering its own proprietary neural/CFR hybrid solver, GTO Wizard AI, which achieved a 0.21\% pot Nash distance against PioSOLVER and defeated ACPC champion Slumbot at 19\text{ bb}/100 over 150,000 hands. Its terms strictly prohibit scraping or redistributing its database. Similarly, Ollin Solver (https://ollinlabs.com/) built an in-house real-time solver engine from scratch to power its coaching interface without third-party licensing constraints, while NashPoint (https://nashpoint.poker/) explicitly pairs an open-source solver engine with AI opponent modeling to avoid commercial solver royalties.

Industry Precedents: How Existing Apps Ship Solver Data

Empirical observation of the commercial poker software ecosystem confirms how these licensing boundaries operate in practice. Multiple commercial mobile and desktop applications openly advertise that their shipped strategy libraries were computed offline using MonkerSolver or HRC. For example, *Poker Preflop Trainer PRO* (developed by WEB CLAP, K.K. on the Apple App Store and Google Play Store) and *PLO Preflop Trainer* (on the Microsoft Store) state verbatim in their commercial store listings: *"The solution is computed in MonkerSolver."*. Likewise, *PreflopWhiz* (https://www.preflopwhiz.com/plans) markets subscription access to *"Monkersolver-powered GTO ranges"*, and numerous tournament study apps ship HRC Pro ranges under the explicit commercial redistribution clause of the HRC v4.0+ EULA [Judgement].
In the open-source ecosystem, browser and desktop applications such as WASM Postflop (https://wasm-postflop.pages.dev/) and Desktop Postflop compile and distribute the b-inary/postflop-solver Rust engine directly to end users under AGPL-3.0, while hybrid training platforms such as NashPoint leverage open-source solver engines on the backend to deliver GTO accuracy without paying PioSOLVER's €5,000/month enterprise fee. Conversely, commercial training tools that ship branded Simple Postflop or PioSOLVER solution packs either

execute formal B2B partnerships—such as Simple GTO Trainer's partnership with GrinderLab—or limit their software integration to a local UPI wrapper that requires the user to own a personal licensed copy of PioSOLVER.

Scenario Comparison: Riverline Closed Source vs. Riverline Open Source

Because the owner of Riverline has not yet committed to a closed-source or open-source model, the architectural trade-offs, usable solvers, financial costs, and legal obligations must be compared directly across both paths.

Scenario 1: Riverline Closed Source (Free App + Optional Paid Tier)

If Riverline remains a proprietary, closed-source application, it still enjoys complete legal access to the best offline solver data pipelines available. For Heads-Up postflop and 6-max bunching-adjusted postflop scenarios, Riverline can run b-inary/postflop-solver offline at zero licensing cost and ship the resulting strategy frequencies and EVs inside both the free tier (a) and the optional paid tier (b), because AGPL-3.0 does not reach numerical outputs [Judgement]. For preflop, multiway preflop, raked cash, and tournament ICM/PKO scenarios, Riverline can subscribe to HRC Pro for $359.90 per year and legally ship all exported calculation results across both the free tier (a) and paid tier (b) under the
0) are also legally usable in a closed-source app, though technically inferior to HRC Pro and b-inary/postflop-solver respectively [Judgement]. Conversely, outputs from GTO+, Simple Postflop, and PioSOLVER remain off-limits without bespoke B2B waivers or PioSOLVER's €5,000/month Business License [Judgement].

Under this closed-source architecture, Riverline cannot compile or bundle the b-inary/postflop-solver or TexasSolver source code directly inside the distributed application binary. However, Riverline can legally embed DeepMind's OpenSpiel (under Apache-2.0) or an in-house clean-room solver, and can also include a UPI socket bridge allowing power users who already own a personal copy of PioSOLVER (or PioSOLVER Free with attribution) to run local solves through Riverline's interface [Judgement]. The total annual licensing cost for this closed-source model is $359.90/year during active HRC Pro simulation generation, with zero source-code disclosure obligations and full intellectual property protection over Riverline's user interface, drill pedagogy, and paid-tier gating [Judgement].

Scenario 2: Riverline Open Source (Licensed under AGPL-3.0-or-later)

A critical second-order insight of this research is that making Riverline open-source changes nothing regarding which commercial solvers' offline data may be shipped [Judgement]. Licensing Riverline under an open-source license does not unlock GTO+, Simple Postflop, or PioSOLVER outputs, because their proprietary EULAs prohibit public redistribution and on-demand lookup services regardless of whether the recipient software is open-source or closed-source [Judgement]. Furthermore, outputs from b-inary/postflop-solver and HRC Pro are already fully cleared for closed-source commercial distribution [Judgement].

What licensing Riverline under AGPL-3.0-or-later actually unlocks is the legal right to compile and bundle the b-inary/postflop-solver Rust engine directly inside the Riverline client (as a native Rust library or WebAssembly module, mirroring desktop-postflop and wasm-postflop).

This empowers Riverline users to run ultra-fast, local on-device custom postflop solves on arbitrary game trees in real time, in addition to studying pre-computed offline data. Notably, choosing a permissive open-source license such as MIT or Apache-2.0 for Riverline would not permit embedding b-inary/postflop-solver; because AGPL-3.0 is a strong copyleft license, the entire combined Riverline application must be licensed under AGPL-3.0-or-later.
In exchange for in-app live solving via b-inary/postflop-solver, AGPL-3.0 obligates Riverline to publish its complete source code and provide a prominent source-code offer to all local and network users. Because any third party is legally entitled under AGPL-3.0 to fork Riverline, strip out client-side paywalls, and redistribute the modified build for free, a paid tier (d) in an open-source Riverline must monetize via server-side value—such as cloud-streamed multi-terabyte HRC Pro and b-inary strategy libraries, cross-device progress synchronization, or voluntary supporter patronage—rather than client-side feature locks [Judgement].

Assessment: Writing an In-House Solver from Published Research

The owner also requested an evaluation of writing a proprietary solver from scratch based solely on published academic literature without copying code from existing AGPL or commercial engines. The mathematical foundations of modern poker solvers—specifically Counterfactual Regret Minimization (Zinkevich et al., 2007), CFR+ (Tammelin, 2014), and Discounted CFR (Noam Brown and Tuomas Sandholm, *Solving Imperfect-Information Games via Discounted Regret Minimization*, AAAI 2019 / arXiv:1809.04040)—are unpatented scientific formulas in the public domain. Under copyright law, mathematical equations and algorithmic procedures (such as discounting positive regrets by t^\alpha / (t^\alpha + 1), negative regrets by t^\beta / (t^\beta + 1), and averaging strategies weighted by t^\gamma) are not copyrightable, meaning a clean-room implementation written from academic papers is 100\% owned by Riverline and can be kept closed-source [Judgement].
However, the engineering effort varies dramatically across game-tree complexity, as summarized in the table below.

Solver Development Scope	Estimated Engineering Effort (1 Senior Systems Engineer)	Key Algorithmic & Systems Engineering Challenges	Expected Quality vs. Existing Engines
Phase 1: River & Turn Subgame Solver	**2 to 4 Weeks** [Judgement]	Implementing a lookup-table 7-card hand evaluator, basic game-tree builder, and vanilla DCFR traversal loop in Rust or C++.	**100% Exact Match.** Turn and river trees are small enough that unoptimized DCFR converges in milliseconds to seconds [Judgement].
Phase 2: Production Heads-Up Flop Solver	**6 to 12 Months** [Judgement]	Requires exact suit isomorphism compression across 2,401 turn/river runouts, SIMD vectorization (AVX2/NEON) over	**High Risk of Slower Speed / Higher RAM.** Matching b-inary/postflop-solver's hand-tuned assembly and SIMD throughput requires months of

Solver Development Scope	Estimated Engineering Effort (1 Senior Systems Engineer)	Key Algorithmic & Systems Engineering Challenges	Expected Quality vs. Existing Engines
		1,326-combo arrays with O(1) card-removal masking, 16-bit integer node quantization, custom arena memory allocators, and floating-point underflow stabilization.	low-level profiling [Judgement].
Phase 3: Multiway & Preflop ICM Solver	**18 to 36+ Months** [Judgement]	Requires k-means / earth-mover-distance postflop information abstraction (up to 16,384 buckets), Variance-Reduced Monte Carlo CFR (VR-MCCFR), and multi-player Malmuth-Harville ICM terminal tree evaluators.	**Inferior to HRC Pro for Years.** Abstraction bucketing errors easily produce exploitable preflop frequencies without years of empirical tuning [Judgement].

From a strategic standpoint, building a full flop or preflop solver from scratch at launch is an unnecessary diversion of engineering capital [Judgement]. Because b-inary/postflop-solver already outperforms PioSOLVER and GTO+ on flop trees and its offline outputs can be shipped legally in a closed-source application for $0, spending six to twelve months re-implementing a flop solver provides zero benefit for an offline-data app [Judgement]. Only if Riverline later decides it must embed a closed-source live flop solver directly inside the client—or wants an instant 2-to-4-week clean-room Turn/River recalculator so shipped flop saves can remain tiny on disk (mirroring GTO+ and Simple Postflop compact saves)—does writing custom clean-room DCFR code become advantageous [Judgement].

Final Recommendation and Actionable Roadmap

Based on the twin priorities of uncompromising solver quality and bulletproof legal compliance, the optimal solver and licensing strategy for Riverline combines the single best postflop engine with the single best preflop/multiway engine in an offline data generation pipeline [Judgement].
For all Heads-Up postflop and 6-max bunching-adjusted postflop scenarios, Riverline should run **b-inary/postflop-solver** offline on internal hardware. It utilizes state-of-the-art Discounted CFR (\gamma=3.0), surpasses PioSOLVER and GTO+ in execution speed, matches their equilibrium accuracy down to 0.1\% exploitability, and uniquely calculates exact card bunching for up to four folded players. The developer should export the resulting numerical strategies (frequencies, EVs, and equities) into Riverline's own clean binary or database schema, ensuring zero source code from the postflop-solver crate is included in the shipped data files [Judgement].
For all preflop, multiway, raked cash game, and tournament (ICM and PKO) scenarios, Riverline should purchase a standard **HRC Pro (HoldemResources Calculator v4.0+)** license at

$359.90 per year. HRC Pro delivers industry-leading 16,384-bucket postflop equity realization and unlimited tree sizes, and its v4.0+ EULA (https://www.holdemresources.net/legal/eula/hrc_v4) explicitly grants the licensee permission to *"distribute, openly share, or use the results of authorized calculations for commercial purposes"* when exported via HRC's native export functions. Conversely, Riverline should strictly avoid shipping data generated by GTO+ (barred by its personal non-commercial EULA), Simple Postflop (barred by Section 2 of its ToS without written permission), and PioSOLVER (barred from on-demand result lookup services without a €5,000/month Business License), and should skip TexasSolver due to its documented benchmark accuracy discrepancies and contradictory README licensing claims [Judgement].

This optimal solver stack implies that **Riverline can and should remain Closed Source** (or adopt a proprietary application layer) if its primary plan is shipping pre-computed strategy data offline [Judgement]. Because neither AGPL-3.0 (for b-inary/postflop-solver outputs) nor the HRC v4.0+ EULA requires the recipient application to be open-source when distributing calculation results, keeping Riverline closed-source is 100\% legal across both the free launch tier (a) and any future paid subscription or supporter tier (b) [Judgement]. If Riverline later desires live, on-device custom postflop solving, it can either write a lightweight clean-room DCFR turn/river solver from Brown and Sandholm (2018) in two to four weeks, allow users to connect an external local solver binary (Desktop Postflop or a user-owned PioSOLVER installation) over a standard inter-process communication (IPC/UPI) pipe, or transition the client to AGPL-3.0-or-later and monetize via cloud-streamed strategy packs [Judgement]. Prior to launching the paid tier, qualified intellectual property counsel should briefly verify that Riverline's custom data serialization format contains strictly numerical arrays without embedded AGPL code, and that dated HRC Pro license invoices and export logs are archived to document full compliance with the Austrian-law HRC v4.0+ EULA [Judgement].

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